Supplementary Information

Supplementary text

S1 Text. Supplementary Methods **2**
MR model and summary statistics . . . 2
The NOME assumption . . . 2
Inverse-variance weighted method . . . 2
Consistency of the IVW estimator . . . 3
MR-Egger regression and the InSIDE assumption . . . 3
Correlated variants . . . 5
Multiplicative random effects and overdispersion . . . 5
Truncated eigendecomposition for the GLS estimators . . . 6

Supplementary figures

S1 Null-SNP diagnostics in Bn, CD4 T cells, and CD8 T cells . . . 8
S2 MR-Egger estimates with and without allele orientation in Bn, CD4 T cells, and CD8 T cells 9
S3 Overlap and effect directions of significant links in Bn and CD8 T cells . . . 10
S4 Sharing of significant links among Bn, CD4 T cells, and CD8 T cells for each method . . . 11
S5 Overlap of MR-only links with regulatory references compared with all candidate peaks . . . 12
S6.1 Pairwise relationships among the five methods with an effect estimate in Bn . . . 14
S6.2 Pairwise relationships among the five methods with an effect estimate in CD4 T cells . . . 15
S6.3 Pairwise relationships among the five methods with an effect estimate in CD8 T cells . . . 16
S7.1 Genome-wide association signals of the six methods in Bn . . . 18
S7.2 Genome-wide association signals of the six methods in CD4 T cells . . . 19
S7.3 Genome-wide association signals of the six methods in CD8 T cells . . . 20
S8 Overlap of significant links with the CIMA regulatory network in Bn and CD8 T cells . . . 21

Supplementary tables

S1 Pooled hypergeometric enrichment p-values across regulatory references. . . . 22
S2 Pooled regulatory-reference fold enrichment across cell types and methods. . . . 23
S3 Data overview for the three cell types analyzed. . . . 24
S4 GWAS used in the S-LDSC analysis (Fig 4A) and in the GWAS-overlap tracks (Fig 4B). . . . 25

S1 Text. Supplementary Methods

MR model and summary statistics

The expressions in this section are given for uncorrelated variants, for exposition. All reported estimates use the correlated-variant (GLS) versions described in the main Methods, which reduce to these expressions when $\boldsymbol{R} = \boldsymbol{I}$.

To connect individual-level associations with the summary statistics used in MR, consider a peak–gene pair with J candidate instruments. Let $G_{ij} \in \{0, 1, 2\}$ denote the number of copies of the effect allele of SNP j carried by individual i, where $j = 1, \ldots, J$ and $i = 1, \ldots, N$, and let X_i and Y_i denote the corresponding accessibility and expression measurements. For each SNP separately, the marginal associations can be written as

$$X_i = \mu_{Xj} + \gamma_j G_{ij} + \varepsilon_{ij}^X,$$
$$Y_i = \mu_{Yj} + \Gamma_j G_{ij} + \varepsilon_{ij}^Y,$$

where μ_{Xj} and μ_{Yj} are intercepts, and γ_j and Γ_j are the population-level SNP–exposure and SNP–outcome associations, respectively. Covariate terms are omitted for simplicity. The two associations are related by

$$\Gamma_j = \beta\gamma_j + \alpha_j, \tag{1}$$

where β is the causal effect of the exposure on the outcome and α_j is the direct (pleiotropic) effect of SNP j on the outcome that is not mediated by the exposure. SNP j is a valid instrument only if $\alpha_j = 0$ [1].

The NOME assumption

Both IVW and MR-Egger regress the estimated SNP–outcome associations $\hat{\Gamma}_j$ on the estimated SNP–exposure associations $\hat{\gamma}_j$ and treat the $\hat{\gamma}_j$ as fixed regressors. This is the NO Measurement Error (NOME) assumption: the sampling error of $\hat{\gamma}_j$ (standard error s_{Xj}) is assumed to be negligible, so that only the uncertainty in $\hat{\Gamma}_j$ enters the weights and the residual variance. When the SNP–exposure associations are imprecise, NOME is violated and the regressor is measured with error. MR-Egger is then subject to weak-instrument bias, which can be assessed with the I^2 statistic proposed by Bowden et al.; values close to one indicate that the MR-Egger estimate does not suffer from weak-instrument bias [1] [CITE: Bowden et al., I^2 statistic for MR-Egger]. All instruments in this study had caQTL nominal $P < 10^{-8}$ (main text, Methods, "Data Preprocessing").

Inverse-variance weighted method

For uncorrelated variants, the IVW method combines the SNP-specific ratio estimates $\hat{\beta}_j = \hat{\Gamma}_j/\hat{\gamma}_j$ into a single estimate of the causal effect. Under NOME and conditional on the $\hat{\gamma}_j$, IVW is a weighted linear regression through the origin [1, 2]:

$$\hat{\Gamma}_j = \beta\hat{\gamma}_j + \varepsilon_j, \qquad \varepsilon_j \overset{\text{ind}}{\sim} \mathcal{N}(0, \phi s_{Yj}^2), \quad j = 1, \ldots, J, \tag{2}$$

where s_{Yj} is the standard error of $\hat{\Gamma}_j$ and ϕ is a residual variance multiplier. The fixed-effect model sets $\phi = 1$, whereas the multiplicative random-effects model estimates ϕ to allow for overdispersion (see "Multiplicative random effects and overdispersion" below). IVW can accommodate balanced pleiotropy with $E(\alpha_j \mid \gamma_j) = 0$ [2].

The IVW estimate minimizes the weighted residual sum of squares

$$Q(\beta) = \sum_{j=1}^{J} \frac{(\hat{\Gamma}_j - \beta\hat{\gamma}_j)^2}{s_{Yj}^2}, \tag{3}$$

which gives

$$\hat{\beta}_{\mathrm{IVW}} = \frac{\sum_{j=1}^{J} s_{Yj}^{-2}\hat{\gamma}_j\hat{\Gamma}_j}{\sum_{j=1}^{J} s_{Yj}^{-2}\hat{\gamma}_j^2} = \frac{\sum_{j=1}^{J} w_j\hat{\beta}_j}{\sum_{j=1}^{J} w_j}, \tag{4}$$

where $w_j = \hat{\gamma}_j^2/s_{Yj}^2$. Thus, the IVW estimate is a weighted average of the SNP-specific ratio estimates, with weights equal to their inverse first-order variances [1]. Its conditional variance is

$$\mathrm{Var}(\hat{\beta}_{\mathrm{IVW}} \mid \hat{\boldsymbol{\gamma}}) = \frac{\phi}{\sum_{j=1}^{J} s_{Yj}^{-2}\hat{\gamma}_j^2}, \tag{5}$$

where $\hat{\boldsymbol{\gamma}}$ is the vector of estimated SNP–exposure associations, and the residual dispersion is estimated by

$$\hat{\phi} = \frac{Q(\hat{\beta}_{\mathrm{IVW}})}{J-1}. \tag{6}$$

Consistency of the IVW estimator

Substituting $\Gamma_j = \beta\gamma_j + \alpha_j$, the ratio estimate of SNP j converges, as the sample size N increases, to

$$\frac{\hat{\Gamma}_j}{\hat{\gamma}_j} \xrightarrow{N\to\infty} \frac{\Gamma_j}{\gamma_j} = \beta + \frac{\alpha_j}{\gamma_j}. \tag{7}$$

The ratio estimate of a valid instrument ($\alpha_j = 0$) is therefore consistent for β, and so is the IVW estimate, a weighted average of the ratio estimates, when all instruments are valid. More generally,

$$\hat{\beta}_{\mathrm{IVW}} \xrightarrow{N\to\infty} \beta + \frac{\sum_{j=1}^{J} s_{Yj}^{-2}\alpha_j\gamma_j}{\sum_{j=1}^{J} s_{Yj}^{-2}\gamma_j^2}, \tag{8}$$

so the asymptotic bias of the IVW estimate is zero if $\sum_j s_{Yj}^{-2}\alpha_j\gamma_j = 0$; otherwise, the IVW estimate is biased [1].

MR-Egger regression and the InSIDE assumption

MR-Egger extends the zero-intercept IVW regression by estimating an intercept, thereby allowing for a nonzero average pleiotropic effect. For uncorrelated SNPs, and treating the estimated SNP–exposure associations as fixed under NOME, the working model is

$$\hat{\Gamma}_j = \alpha_0 + \beta\hat{\gamma}_j + \varepsilon_{\mathrm{Egger},j}, \qquad \varepsilon_{\mathrm{Egger},j} \overset{\mathrm{ind}}{\sim} \mathcal{N}(0, \phi_{\mathrm{Egger}} s_{Yj}^2), \tag{9}$$

where α_0 is the intercept and ϕ_{Egger} is a residual variance multiplier [1].

Allele orientation. Changing the coded allele of a SNP reverses the signs of both $\hat{\gamma}_j$ and $\hat{\Gamma}_j$. This leaves the IVW regression through the origin unchanged, but in MR-Egger it changes the definition of α_j, and therefore the intercept, the slope and the InSIDE assumption itself. Both associations are therefore oriented to the exposure-increasing allele, so that $\hat{\gamma}_j > 0$ for all j; MR-Egger then does not depend on the original allele coding, and directional pleiotropy is defined with respect to the exposure-increasing allele [1] (main text, Methods, "Allele orientation").

Estimation. Let $v_j = s_{Yj}^{-2}$ denote the regression weights and define the weighted means

$$\bar{\gamma}_v = \frac{\sum_{j=1}^{J} v_j\hat{\gamma}_j}{\sum_{j=1}^{J} v_j}, \qquad \bar{\Gamma}_v = \frac{\sum_{j=1}^{J} v_j\hat{\Gamma}_j}{\sum_{j=1}^{J} v_j}. \tag{10}$$

The weighted least-squares estimates minimize

$$Q_{\text{Egger}}(\alpha_0, \beta) = \sum_{j=1}^{J} v_j(\hat{\Gamma}_j - \alpha_0 - \beta\hat{\gamma}_j)^2. \tag{11}$$

Writing $S_{\gamma\gamma} = \sum_{j=1}^{J} v_j(\hat{\gamma}_j - \bar{\gamma}_v)^2$, the estimates are

$$\begin{aligned} \hat{\beta}_{\text{Egger}} &= \frac{\sum_{j=1}^{J} v_j(\hat{\gamma}_j - \bar{\gamma}_v)(\hat{\Gamma}_j - \bar{\Gamma}_v)}{S_{\gamma\gamma}}, \\ \hat{\alpha}_0 &= \bar{\Gamma}_v - \hat{\beta}_{\text{Egger}}\bar{\gamma}_v. \end{aligned} \tag{12}$$

Consistency and the InSIDE assumption. The MR-Egger slope is the weighted covariance between $\hat{\Gamma}_j$ and $\hat{\gamma}_j$ divided by the weighted variance of $\hat{\gamma}_j$, with weights v_j. For a fixed set of J instruments, as N increases,

$$\hat{\beta}_{\text{Egger}} = \frac{\text{cov}_v(\hat{\boldsymbol{\Gamma}}, \hat{\boldsymbol{\gamma}})}{\text{var}_v(\hat{\boldsymbol{\gamma}})} \xrightarrow{N\to\infty} \frac{\text{cov}_v(\boldsymbol{\Gamma}, \boldsymbol{\gamma})}{\text{var}_v(\boldsymbol{\gamma})} = \beta + \frac{\text{cov}_v(\boldsymbol{\alpha}, \boldsymbol{\gamma})}{\text{var}_v(\boldsymbol{\gamma})}, \tag{13}$$

where bold symbols denote vectors across instruments and, for any two such vectors $\boldsymbol{a}$ and $\boldsymbol{b}$,

$$\text{cov}_v(\boldsymbol{a}, \boldsymbol{b}) = \frac{\sum_{j=1}^{J} v_j(a_j - \bar{a}_v)(b_j - \bar{b}_v)}{\sum_{j=1}^{J} v_j}, \qquad \text{var}_v(\boldsymbol{a}) = \text{cov}_v(\boldsymbol{a}, \boldsymbol{a}), \tag{14}$$

with $\bar{a}_v$ and $\bar{b}_v$ the v-weighted means. The MR-Egger slope is therefore consistent for β if $\text{cov}_v(\boldsymbol{\alpha}, \boldsymbol{\gamma}) = 0$, that is, if across the instruments in the analysis the pleiotropic effects are uncorrelated with the instrument strengths. This is the Instrument Strength Independent of Direct Effect (InSIDE) assumption [1, 2].

Under InSIDE, the intercept converges to the weighted average pleiotropic effect $\bar{\alpha}_v = \sum_j v_j\alpha_j / \sum_j v_j$. When pleiotropy is balanced ($\bar{\alpha}_v = 0$) and InSIDE holds, the intercept is zero, the MR-Egger and IVW estimates coincide, and the IVW estimate is also consistent, because the two conditions together imply that the IVW bias term $\sum_j s_{Xj}^{-2}\alpha_j\gamma_j$ is zero. Conversely, a nonzero intercept indicates that the average pleiotropic effect differs from zero (directional pleiotropy), that InSIDE is violated, or both [1].

The condition above is a finite-sample version of InSIDE: the weighted covariance is zero for the particular set of instruments analyzed, and consistency holds as N increases with J fixed. In the original description of MR-Egger [3], InSIDE was stated as independence between the distribution of pleiotropic effects and the distribution of SNP–exposure associations, with the instruments regarded as a sample from a population of variants; under this population version, consistency also requires the number of instruments to tend to infinity. Balanced and directional pleiotropy are defined correspondingly, as a zero or nonzero weighted mean of the distribution of pleiotropic effects (population version) or of the pleiotropic effects of the instruments analyzed (finite-sample version) [1].

InSIDE is plausible when the pleiotropic effects act through variables that do not confound the exposure–outcome relationship. It is likely to be violated when the pleiotropic effects act through confounders: a variant with a larger effect on a confounder then has both a larger association with the exposure and a larger pleiotropic effect on the outcome, so that instrument strength and pleiotropy are correlated [1].

Precision. Under the working model, the conditional slope variance is

$$\text{Var}(\hat{\beta}_{\text{Egger}} \mid \hat{\boldsymbol{\gamma}}) = \frac{\phi_{\text{Egger}}}{S_{\gamma\gamma}}. \tag{15}$$

The precision of the slope therefore depends on the variation in the SNP–exposure associations, not only on their overall magnitude. When the exposure associations are similar across SNPs, the MR-Egger estimate can be imprecise even if each association is strong [1]. Because the regression estimates both an intercept and

a slope, the residual dispersion is estimated with $J-2$ degrees of freedom:

$$\hat{\phi}_{\text{Egger}} = \frac{Q_{\text{Egger}}(\hat{\alpha}_0, \hat{\beta}_{\text{Egger}})}{J-2}. \tag{16}$$

Correlated variants

For correlated instruments, both regressions are fitted by generalized weighted least squares with weight matrix $\boldsymbol{\Psi} = \boldsymbol{\Sigma}^{-1}$, where $\boldsymbol{\Sigma} = \boldsymbol{D}_Y \boldsymbol{R} \boldsymbol{D}_Y$ (main text, Methods, "Causal IVW"). For a fixed set of J instruments, the causal IVW estimate converges to

$$\hat{\beta}_{\text{GLS-IVW}} \xrightarrow{N\to\infty} \beta + \frac{\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\boldsymbol{\alpha}}{\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\boldsymbol{\gamma}}, \tag{17}$$

and the causal Egger slope converges to

$$\hat{\beta}_{\text{GLS-Egger}} \xrightarrow{N\to\infty} \frac{(\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\boldsymbol{\Gamma})(\mathbf{1}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1}) - (\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1})(\boldsymbol{\Gamma}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1})}{(\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\boldsymbol{\gamma})(\mathbf{1}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1}) - (\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1})^2} = \beta + \frac{(\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\boldsymbol{\alpha})(\mathbf{1}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1}) - (\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1})(\boldsymbol{\alpha}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1})}{(\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\boldsymbol{\gamma})(\mathbf{1}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1}) - (\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1})^2}, \tag{18}$$

where the second equality uses $\boldsymbol{\Gamma} = \beta\boldsymbol{\gamma} + \boldsymbol{\alpha}$. The correlated-variant InSIDE condition is therefore

$$(\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\boldsymbol{\alpha})(\mathbf{1}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1}) - (\boldsymbol{\gamma}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1})(\boldsymbol{\alpha}^\mathsf{T}\boldsymbol{\Psi}\mathbf{1}) = 0. \tag{19}$$

When the variants are uncorrelated, $\boldsymbol{\Psi} = \text{diag}(v_1, \ldots, v_J)$, and this condition reduces to $\text{cov}_v(\boldsymbol{\alpha}, \boldsymbol{\gamma}) = 0$ [1]. In our implementation, $\boldsymbol{\Sigma}^{-1}$ is replaced by the truncated pseudoinverse $\widetilde{\boldsymbol{\Sigma}}^{+}$ (see "Truncated eigendecomposition for the GLS estimators" below); the expressions above hold with $\boldsymbol{\Psi} = \widetilde{\boldsymbol{\Sigma}}^{+}$ because they depend on the weighting only through $\boldsymbol{\Psi}$. Burgess and Thompson noted that this criterion is unlikely to be satisfied if all variants are mutually correlated, because correlation between variants is likely to induce correlation between their exposure associations and pleiotropic effects [1].

Multiplicative random effects and overdispersion

Rationale. The fixed-effect model ($\phi = 1$) assumes that the residuals reflect only the sampling error of the $\hat{\Gamma}_j$, which holds when all instruments estimate the same causal effect. Pleiotropic effects ($\alpha_j \neq 0$) add variation among the SNP-specific ratio estimates beyond sampling error. This heterogeneity appears as overdispersion: the residual variance exceeds that implied by the $s^2_{Y_j}$ (or by $\boldsymbol{\Sigma}$ for correlated variants). In our setting, many instruments may affect gene expression through pathways other than accessibility at the tested peak, so heterogeneity among the ratio estimates is expected. Ignoring it would give standard errors that are too small and inflate the type I error. A random-effects model that allows for multiplicative overdispersion is therefore preferred for IVW when pleiotropy is suspected, and should always be used for MR-Egger [1].

Procedure. For both methods, we proceeded in four steps.

1. Fit the regression. The point estimates do not depend on ϕ and are therefore identical under the fixed-effect and random-effects models.
2. Estimate the residual dispersion from the residual quadratic form Q at the fitted values, $\hat{\phi} = Q/(J-p)$, with $p = 1$ for IVW and $p = 2$ for MR-Egger. In the absence of heterogeneity, $\hat{\phi}$ is expected to be close to one; $\hat{\phi} > 1$ indicates overdispersion.
3. Set $\widetilde{\phi} = \max(1, \hat{\phi})$. For uncorrelated variants, underdispersion ($\hat{\phi} < 1$) has no plausible biological mechanism and may arise by chance; scaling by $\hat{\phi} < 1$ would make the random-effects estimate more precise than the fixed-effect estimate, so standard errors are inflated under overdispersion but never reduced [1].
4. Multiply the fixed-effect standard errors by $\widetilde{\phi}^{1/2}$ and compare the resulting test statistic with a t distribution with $J-p$ degrees of freedom.

For IVW, the resulting standard error is

$$\widehat{\mathrm{SE}}(\hat{\beta}_{\mathrm{IVW}}) = \left\{ \frac{\widetilde{\phi}}{\sum_{j=1}^{J} s_{Yj}^{-2} \hat{\gamma}_j^2} \right\}^{1/2}, \tag{20}$$

and inference is based on

$$T_{\mathrm{IVW}} = \frac{\hat{\beta}_{\mathrm{IVW}}}{\widehat{\mathrm{SE}}(\hat{\beta}_{\mathrm{IVW}})} \tag{21}$$

with a t_{J-1} reference distribution. For MR-Egger, with $J > 2$ and $S_{\gamma\gamma} > 0$, $\widetilde{\phi}_{\mathrm{Egger}} = \max(1, \hat{\phi}_{\mathrm{Egger}})$, and the slope and intercept standard errors are

$$\begin{aligned} \widehat{\mathrm{SE}}(\hat{\beta}_{\mathrm{Egger}}) &= \left(\frac{\widetilde{\phi}_{\mathrm{Egger}}}{S_{\gamma\gamma}} \right)^{1/2}, \\ \widehat{\mathrm{SE}}(\hat{\alpha}_0) &= \left\{ \widetilde{\phi}_{\mathrm{Egger}} \left[\frac{1}{\sum_{j=1}^{J} v_j} + \frac{\bar{\gamma}_v^2}{S_{\gamma\gamma}} \right] \right\}^{1/2}. \end{aligned} \tag{22}$$

The slope test evaluates $H_0 : \beta = 0$ using

$$T_{\mathrm{Egger}} = \frac{\hat{\beta}_{\mathrm{Egger}}}{\widehat{\mathrm{SE}}(\hat{\beta}_{\mathrm{Egger}})}, \tag{23}$$

with two-sided P values from a t_{J-2} reference distribution.

The same procedure was applied to causal IVW and causal Egger, with the residual quadratic forms weighted by $\widetilde{\boldsymbol{\Sigma}}^{+}$ and with J equal to the number of instruments after LD filtering (main text, Methods, "Causal IVW" and "Causal Egger"). Pairs with $\hat{\phi}_{\mathrm{IVW}} > 10$ were removed before multiple-testing correction (main text, Methods, "Filtering and multiple testing").

Truncated eigendecomposition for the GLS estimators

Strong LD can make $\boldsymbol{\Sigma}$ nearly singular, so that direct inversion is numerically unstable. Without truncation, the GLS estimate of causal IVW would minimize

$$Q_{\mathrm{GLS}}(\beta) = (\hat{\boldsymbol{\Gamma}} - \beta\hat{\boldsymbol{\gamma}})^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} (\hat{\boldsymbol{\Gamma}} - \beta\hat{\boldsymbol{\gamma}}). \tag{24}$$

We therefore regularized the covariance weighting with a truncated eigendecomposition:

$$\boldsymbol{\Sigma} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\mathsf{T}}, \qquad \boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_J), \tag{25}$$

where $\boldsymbol{V}$ is orthogonal and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_J$. Let $\lambda_\ell^+ = \max(\lambda_\ell, 0)$ and define

$$m_\eta = \min \left\{ m \in \{1, \ldots, J\} : \frac{\sum_{\ell=1}^{m} \lambda_\ell^+}{\sum_{\ell=1}^{J} \lambda_\ell^+} \geq \eta \right\}, \qquad \eta = 0.999. \tag{26}$$

The retained directions are

$$\mathcal{K} = \left\{ \ell : 1 \leq \ell \leq \max(m_\eta, 5),\ \lambda_\ell > 10^{-10}\lambda_1 \right\}, \qquad k = |\mathcal{K}|. \tag{27}$$

That is, at least five directions are retained when $J \geq 5$, and the relative eigenvalue threshold excludes directions with very small or non-positive eigenvalues. Because the eigenvalues are sorted, $\mathcal{K}$ consists of the k leading directions.

We denote the truncated covariance approximation by

$$\widetilde{\boldsymbol{\Sigma}} = \boldsymbol{V}\,\mathrm{diag}(\widetilde{\lambda}_1, \ldots, \widetilde{\lambda}_J)\boldsymbol{V}^{\mathsf{T}}, \qquad \widetilde{\lambda}_\ell = \begin{cases} \lambda_\ell, & \ell \in \mathcal{K}, \\ 0, & \ell \notin \mathcal{K}. \end{cases} \tag{28}$$

By the Eckart–Young theorem, $\widetilde{\boldsymbol{\Sigma}}$ is the best rank-k approximation of $\boldsymbol{\Sigma}$ in both the Frobenius and spectral norms [CITE: Eckart–Young theorem]. When $k < J$, this matrix is singular. We therefore use its Moore–Penrose pseudoinverse,

$$\widetilde{\boldsymbol{\Sigma}}^{+} = \boldsymbol{V}\,\mathrm{diag}(d_1, \ldots, d_J)\boldsymbol{V}^{\mathsf{T}}, \qquad d_\ell = \begin{cases} \lambda_\ell^{-1}, & \ell \in \mathcal{K}, \\ 0, & \ell \notin \mathcal{K}, \end{cases} \tag{29}$$

which inverts the covariance on the retained eigenspace and assigns zero weight to the discarded directions.

With this weighting, causal IVW minimizes

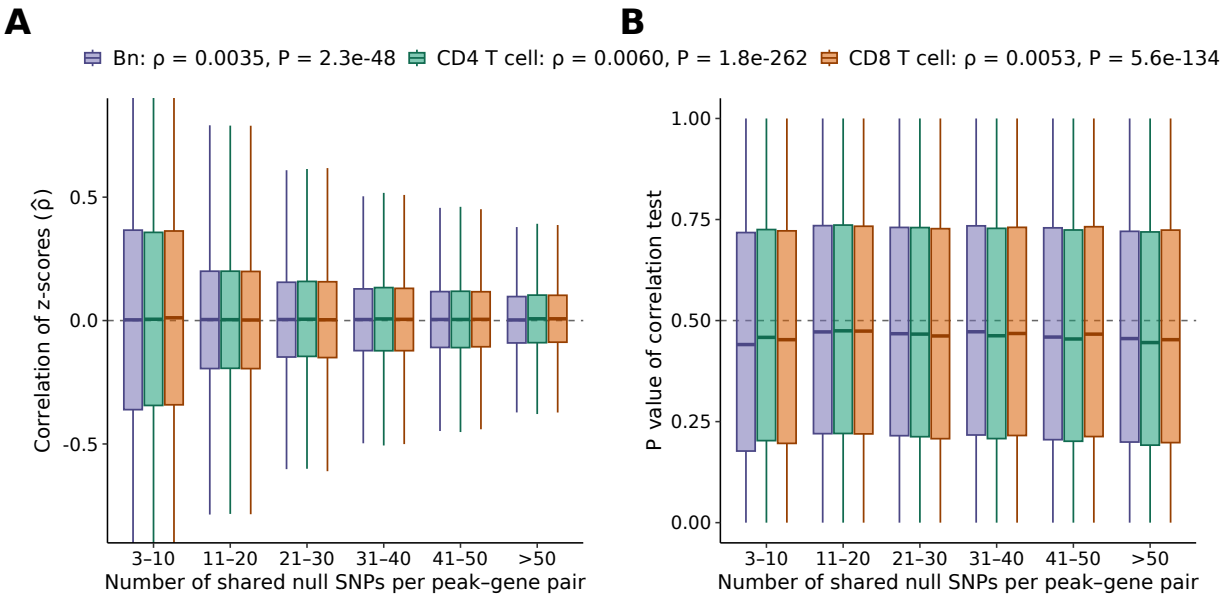
$$Q_{\mathcal{K}}(\beta) = (\hat{\boldsymbol{\Gamma}} - \beta\hat{\boldsymbol{\gamma}})^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}(\hat{\boldsymbol{\Gamma}} - \beta\hat{\boldsymbol{\gamma}}), \tag{30}$$

and causal Egger, with $\boldsymbol{X} = [\mathbf{1}_J, \hat{\boldsymbol{\gamma}}]$ and $\boldsymbol{\theta} = (\alpha_0, \beta)^{\mathsf{T}}$, minimizes

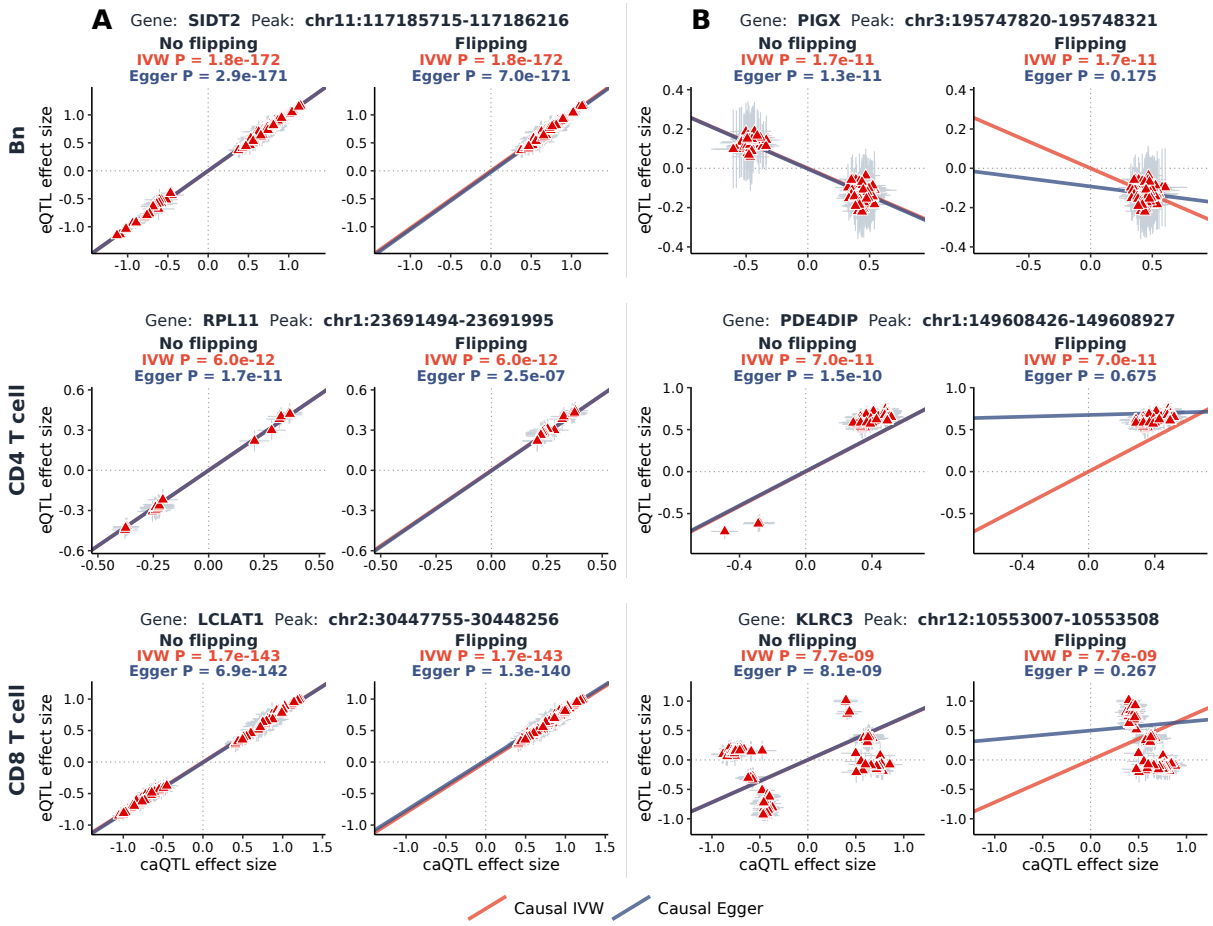
$$Q_{\mathrm{Egger}}(\boldsymbol{\theta}) = (\hat{\boldsymbol{\Gamma}} - \boldsymbol{X}\boldsymbol{\theta})^{\mathsf{T}}\widetilde{\boldsymbol{\Sigma}}^{+}(\hat{\boldsymbol{\Gamma}} - \boldsymbol{X}\boldsymbol{\theta}). \tag{31}$$

The closed-form solutions and their conditional variances are given in the main Methods; the residual quadratic forms at the solutions are the Q used for $\hat{\phi}$.

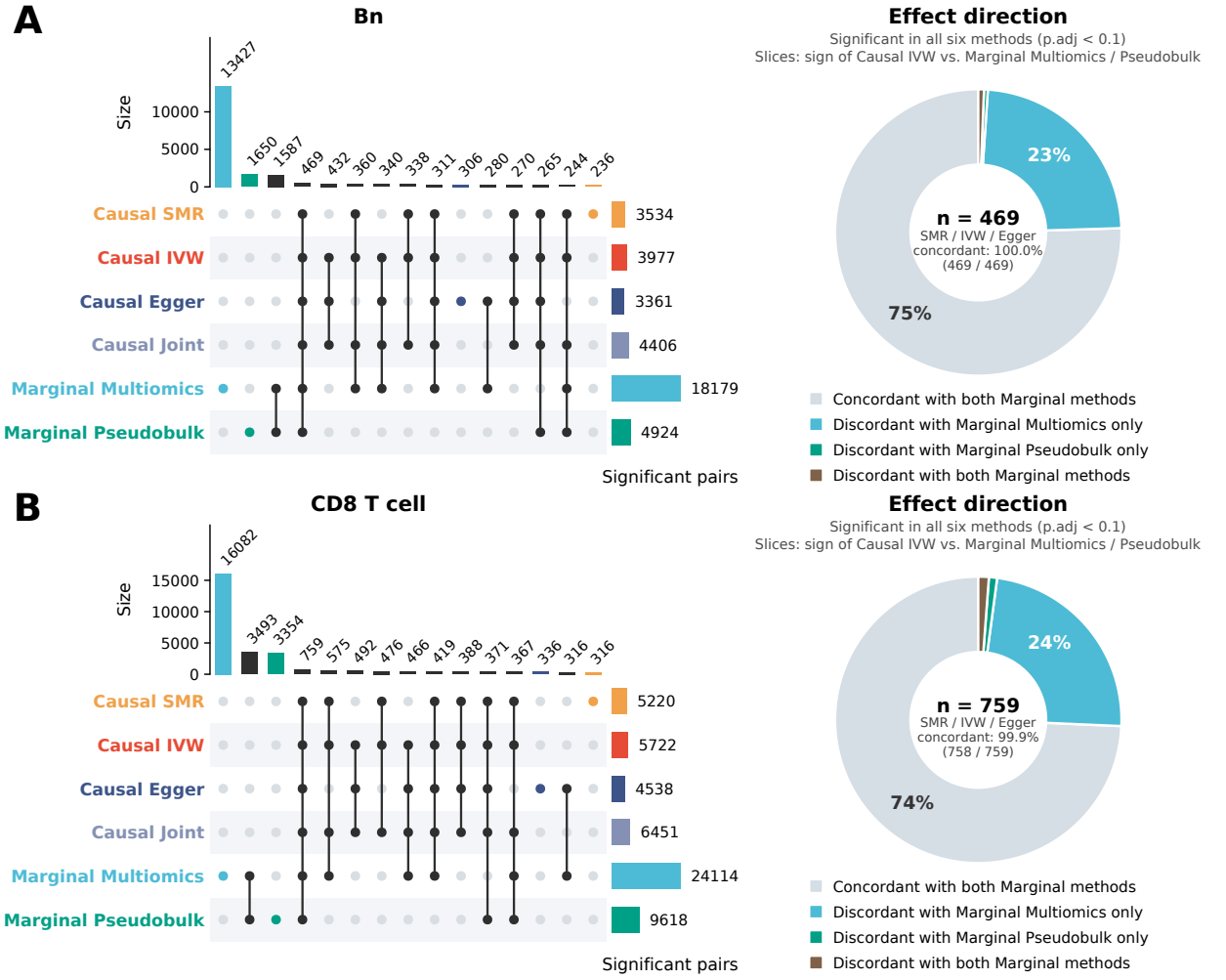
In the implementation, the estimates were computed by symmetric whitening on $\mathcal{K}$, $\boldsymbol{W} = \boldsymbol{V}\,\mathrm{diag}(d_1^{1/2}, \ldots, d_J^{1/2})\boldsymbol{V}^{\mathsf{T}}$, followed by ordinary least squares of $\boldsymbol{W}\hat{\boldsymbol{\Gamma}}$ on $\boldsymbol{W}\hat{\boldsymbol{\gamma}}$ (causal IVW, through the origin) or on $\boldsymbol{W}[\mathbf{1}_J, \hat{\boldsymbol{\gamma}}]$ (causal Egger); this is equivalent to the pseudoinverse form above. The residual degrees of freedom are $J - 1$ (causal IVW) and $J - 2$ (causal Egger), where J is the number of instruments after LD filtering, not the number of retained directions k.



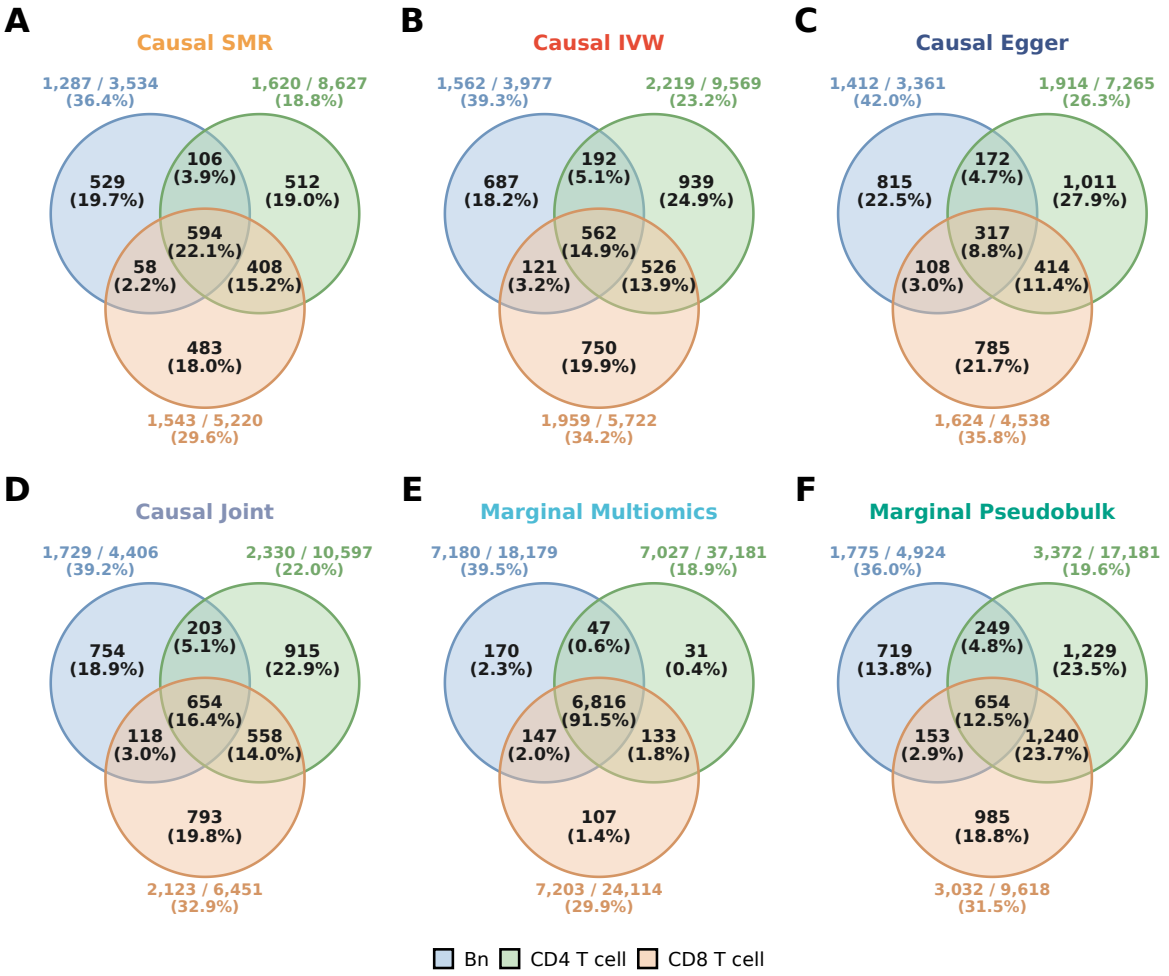
S1 Fig. Null-SNP diagnostics in Bn, CD4 T cells, and CD8 T cells. **A,** Pair-level correlations of caQTL and eQTL z-scores, grouped by the number of shared pruned null SNPs; boxes show the three cell types side by side (Bn, purple; CD4 T cells, green; CD8 T cells, orange). The legend gives the global correlation ρ and its p-value for each cell type. **B,** Corresponding p-values of the pair-level correlation tests. The dashed lines mark zero correlation in A and a p-value of 0.5 in B, the values expected under the null. The analyses included 547,452 (Bn), 1,051,953 (CD4 T cells), and 686,829 (CD8 T cells) peak–gene pairs, based on 17,074,938, 33,263,695, and 21,537,822 SNP–pair observations, respectively. Global summaries pool SNP–pair observations, so a variant may occur in multiple pairs. This diagnostic does not itself establish MR calibration.



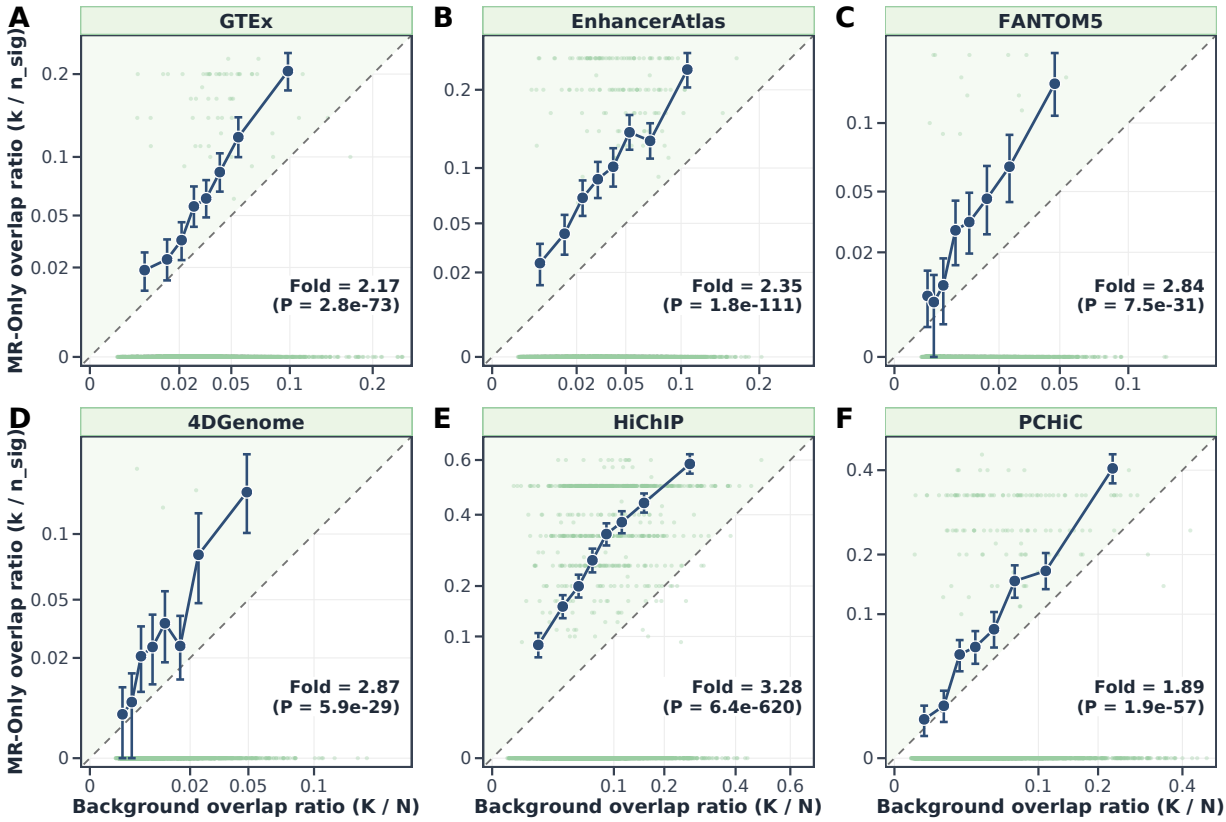
S2 Fig. MR-Egger estimates with and without allele orientation in Bn, CD4 T cells, and CD8 T cells. Rows show Bn, CD4 T cells, and CD8 T cells, and each pair of plots is one peak–gene pair. **A,** pairs whose estimates change little with orientation (SIDT2 in Bn, 194 SNPs; RPL11 in CD4 T cells, 25 SNPs; LCLAT1 in CD8 T cells, 195 SNPs). **B,** pairs whose MR-Egger estimate changes markedly (PIGX in Bn, 136 SNPs; PDE4DIP in CD4 T cells, 74 SNPs; KLRC3 in CD8 T cells, 134 SNPs). For each pair, the left plot (No flipping) uses the original allele coding and the right plot (Flipping) orients the pair so that the caQTL effect is positive, as in the MR analysis; both plots share axis ranges. Red and blue lines are the IVW and MR-Egger fits, and the P values above each plot are the corresponding IVW and MR-Egger P values. Red triangles are SNPs and gray bars are 95% confidence intervals. IVW is unchanged by orientation.



S3 Fig. Overlap and effect directions of significant links in Bn and CD8 T cells. **A,** Bn; **B,** CD8 T cells. In each row, left: UpSet plot of the 15 largest distinct significance combinations across six methods at within-gene BH-adjusted $P_{\text{adj}} < 0.1$; top bars give the size of each combination, and right bars give the number of significant pairs for each method. Right: among links significant in all six methods, the sign of the IVW estimate relative to the signs of the two marginal correlations (categories in the legend below the ring); the center reports the number of such links and the proportion whose SMR, IVW, and Egger estimates agree in sign. "Causal Joint" denotes CCT. The corresponding CD4 T cell comparison is shown in Fig 1B–C.

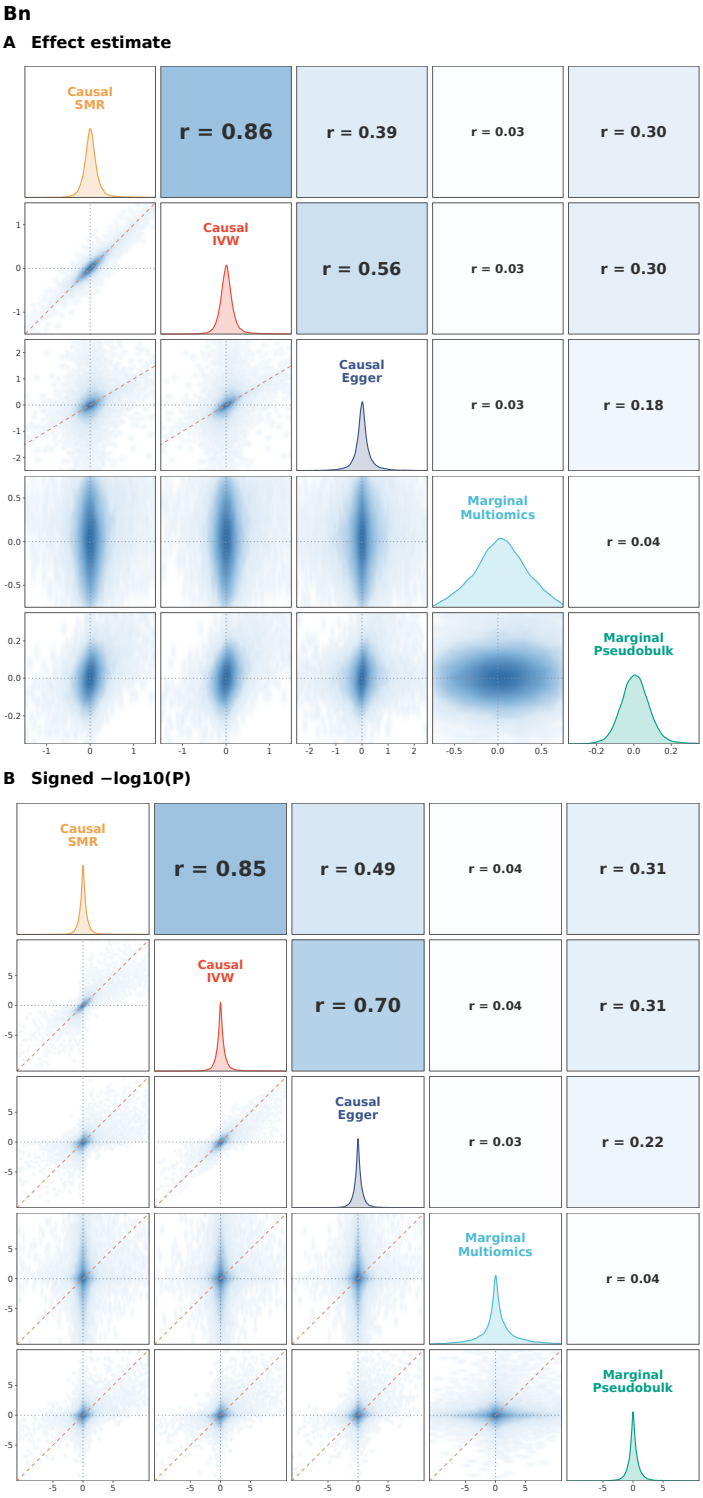


S4 Fig. Sharing of significant links among Bn, CD4 T cells, and CD8 T cells for each method. **A–F,** causal SMR, causal IVW, causal Egger, causal joint, marginal multiomics, and marginal pseudobulk. To compare the same pairs across cell types, each diagram is restricted to the 14,890 candidate peak–gene pairs that were tested in all three cell types (of 40,087, 85,527, and 50,931 pairs tested in Bn, CD4 T cells, and CD8 T cells, respectively). For each method, the circle of a cell type contains the common pairs with $P_{\mathrm{adj}} < 0.1$ in that cell type, where P_{adj} is the within-gene BH-adjusted P value of that cell type's own analysis; pairs were matched by peak and gene. Numbers inside the diagram give the number of links in each region and, in parentheses, its percentage of all links significant in at least one cell type for that method. Labels outside each circle give the number of that cell type's links that fall in the common pairs over the total number of links of that method in that cell type (all pairs tested in that cell type), with the percentage in parentheses. Circle areas are not proportional to set sizes. Marginal multiomics correlations were computed on WNN metacells from an external PBMC dataset without cell-type separation, so a common pair has the same unadjusted P value in all three cell types and differs only in the within-gene BH adjustment.

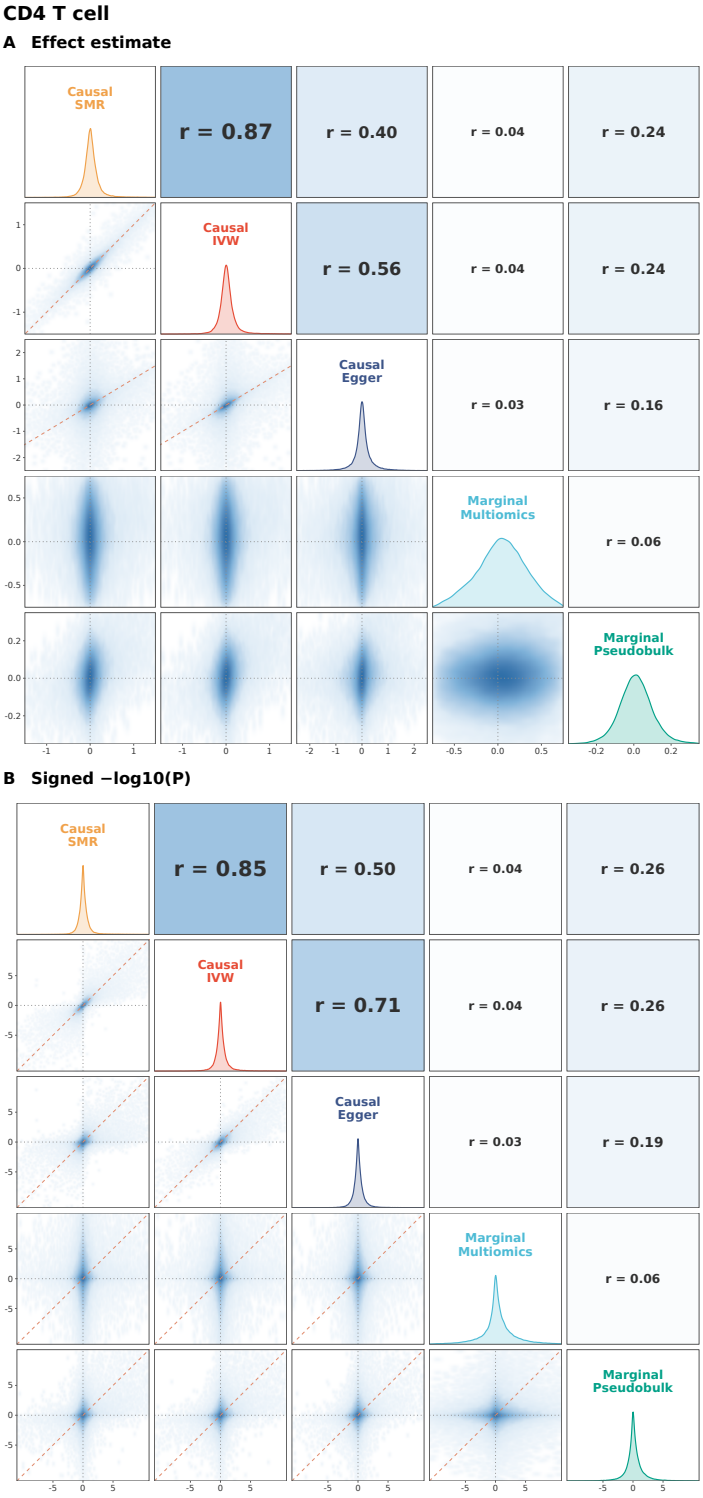


S5 Fig. Overlap of MR-only links with regulatory references compared with all candidate peaks. **A–F,** GTEx, EnhancerAtlas, FANTOM5, 4DGenome, HiChIP, and PCHi-C (PCHiC in the figure). MR-only links ($P_{\text{adj}} < 0.1$ for at least one of causal IVW, causal Egger, and causal SMR, and $P_{\text{adj}} \geq 0.1$ for both marginal methods; see Methods) were pooled across the three cell types. For each gene, N is the number of candidate peaks (union across cell types), K the number of these peaks that overlap the gene's reference regions, n_{sig} the number of peaks of the gene's MR-only links (union across cell types), and k the number of these MR-only peaks that overlap the reference regions. For HiChIP and PCHi-C, the reference regions are the union of the B cell, CD4 T cell, and CD8 T cell interactions. Genes with $K > 0$ and $N > K$ were included. The x-axis is the background overlap ratio K/N and the y-axis the MR-only overlap ratio k/n_{sig}; both axes are on a square-root scale. Light green points are individual genes; genes outside the plotted range are not shown. About 55% of genes have a single MR-only peak, so per-gene ratios take few distinct values and form horizontal bands, including the band at zero. Dark blue points: in each panel, genes were divided into eight groups of equal size by K/N, and each point gives the pooled ratios $\sum K / \sum N$ and $\sum k / \sum n_{\text{sig}}$ of a group. Error bars are 95% bootstrap confidence intervals of $\sum k / \sum n_{\text{sig}}$ (2,000 resamples of genes within each group, percentile method). The dashed line is $y = x$, the overlap expected if MR-only peaks overlapped the references at the background rate; the shaded area above it marks higher overlap among MR-only peaks. The lower right of each panel gives the pooled fold enrichment $(\sum k / \sum n_{\text{sig}}) / (\sum K / \sum N)$ and the P value of a hypergeometric test on the summed counts, computed as in S1 and S2 Tables. Panel E is also shown as Fig 3D of the main text.

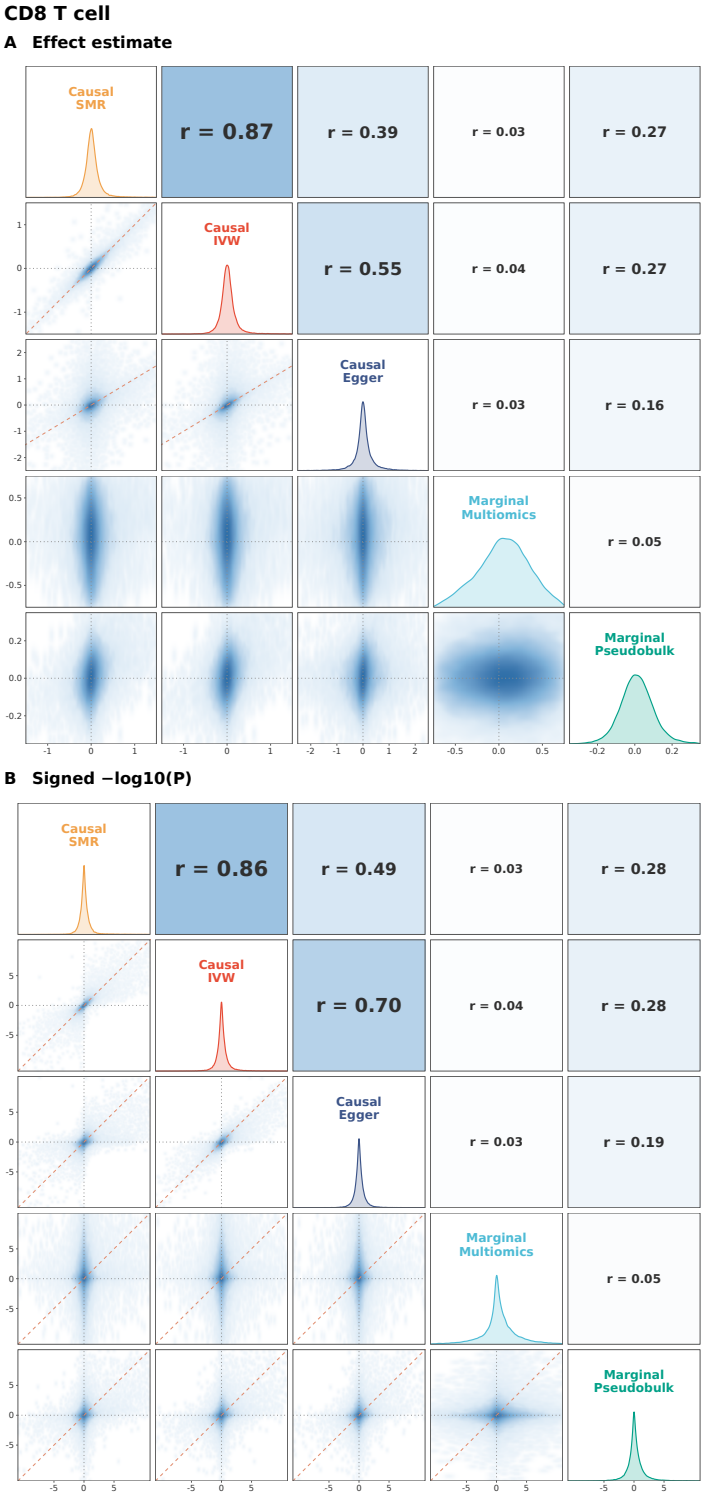
Legend for S6.1–S6.3 Figs. Pairwise relationships among the five methods with an effect estimate in Bn, CD4 T cells, and CD8 T cells. S6.1, S6.2, and S6.3 Figs show Bn, CD4 T cells, and CD8 T cells, respectively, each using all candidate peak–gene pairs tested in that cell type (40,087, 85,527, and 50,931 pairs). Causal joint is not shown because it combines P values and has no effect estimate. **A,** effect estimates: the MR estimate β for causal SMR, causal IVW, and causal Egger, and the Spearman correlation coefficient ρ for marginal multiomics and marginal pseudobulk. **B,** signed $-\log_{10} P$, that is, the sign of the effect estimate multiplied by $-\log_{10} P$. In each matrix, the diagonal gives the method name and the distribution of that method's values across pairs (kernel density; the curve height is rescaled within the panel, so only its shape is informative). Panels below the diagonal show the two-dimensional kernel density of pairs, with the method of the column on the x-axis and the method of the row on the y-axis; color intensity increases with the fourth root of the density so that low-density regions remain visible. Panels above the diagonal give the Spearman correlation r between the two methods across all pairs; text size and background shade increase with r. Dotted lines mark zero. The dashed line is $y = x$; in A it is drawn only between MR methods, whose estimates share units, and in B it is drawn in all panels. Axes are symmetric around zero and cover the 99th percentile of absolute values across the three cell types: in A, each method has its own range (± 1.5 for causal SMR and causal IVW, ± 2.5 for causal Egger, ± 0.75 for marginal multiomics, and ± 0.35 for marginal pseudobulk); in B, all methods share ± 11. Pairs outside these ranges (0.3–1.9% per method and cell type) are omitted from the density panels but included in r. Marginal multiomics correlations were computed on WNN metacells from an external PBMC dataset without cell-type separation.



S6.1 Fig. Pairwise relationships among the five methods with an effect estimate in Bn. Based on the 40,087 candidate peak–gene pairs tested in Bn. **A,** effect estimates; **B,** signed $-\log_{10} P$. See the shared legend for S6.1–S6.3 Figs.

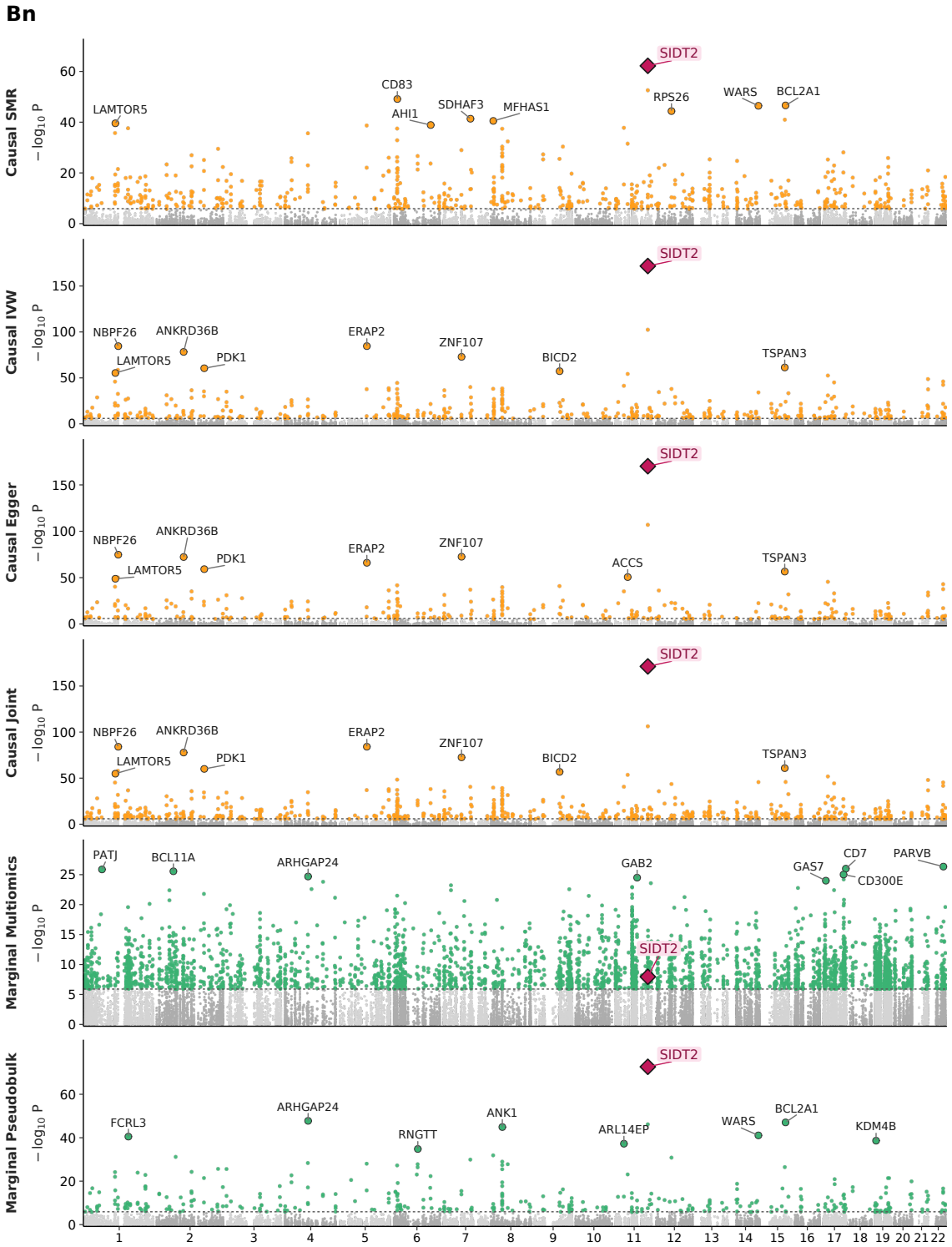


S6.2 Fig. Pairwise relationships among the five methods with an effect estimate in CD4 T cells. Based on the 85,527 candidate peak–gene pairs tested in CD4 T cells. **A,** effect estimates; **B,** signed $-\log_{10} P$. See the shared legend for S6.1–S6.3 Figs.

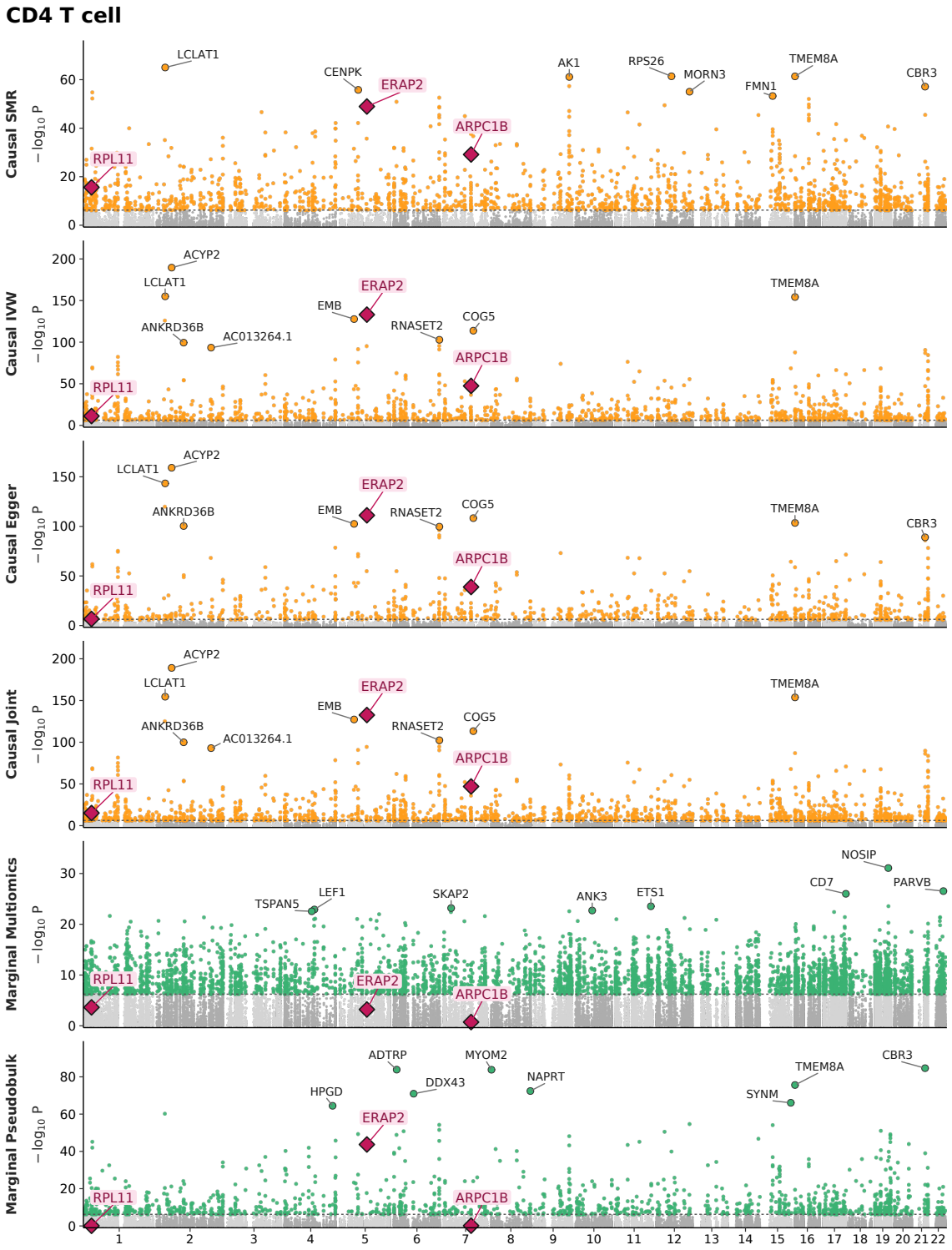


S6.3 Fig. Pairwise relationships among the five methods with an effect estimate in CD8 T cells. Based on the 50,931 candidate peak–gene pairs tested in CD8 T cells. **A,** effect estimates; **B,** signed $-\log_{10} P$. See the shared legend for S6.1–S6.3 Figs.

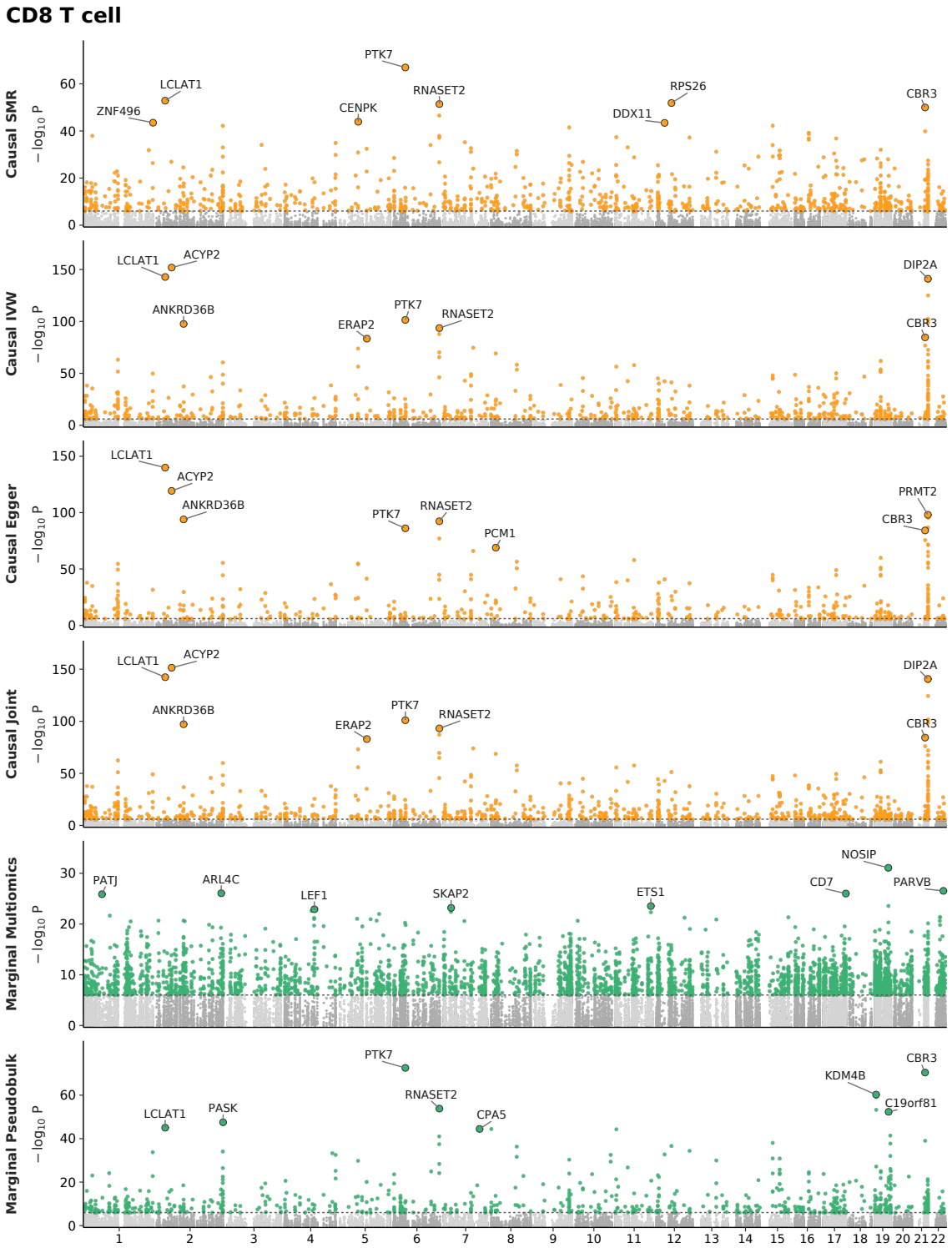
Legend for S7.1–S7.3 Figs. Genome-wide association signals of the six methods in Bn, CD4 T cells, and CD8 T cells. S7.1, S7.2, and S7.3 Figs show Bn, CD4 T cells, and CD8 T cells, respectively, each using all candidate peak–gene pairs tested in that cell type (40,087, 85,527, and 50,931 pairs). Each track is one method (causal SMR, causal IVW, causal Egger, causal joint, marginal multiomics, and marginal pseudobulk). Each point is a pair placed at the center of its peak along chromosomes 1–22 (hg38); the y-axis gives $-\log_{10} P$ on a linear scale with a separate range for each track. The dashed line marks the Bonferroni threshold (0.05 divided by the number of pairs tested in that cell type); pairs above it are colored orange for causal methods and green for marginal methods, and the other pairs are shown in alternating shades of gray by chromosome. In each track, up to eight genes above the threshold are labeled at their strongest pair, with at most one label per 5 Mb on a chromosome. Magenta diamonds mark the example pairs of Fig 2 in the cell type in which they were shown (*SIDT2* in Bn; *RPL11*, *ERAP2*, and *ARPC1B* in CD4 T cells), labeled in every track at their actual position whether or not they pass the threshold. CD8 T cells have no example pair.



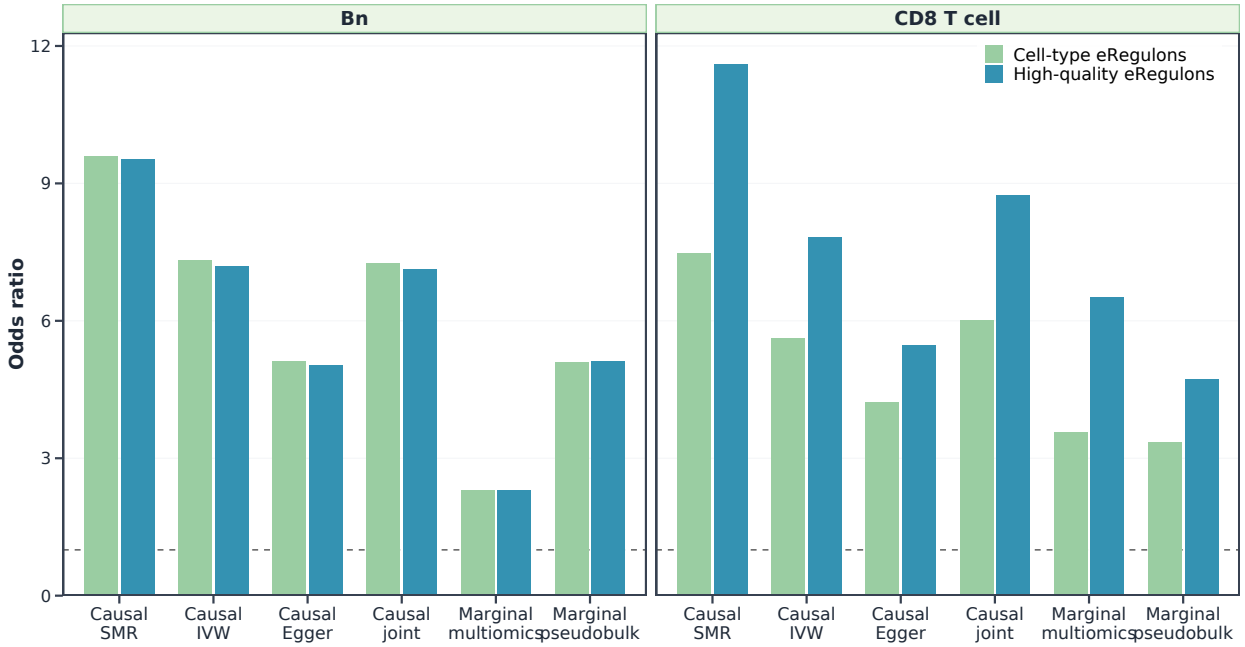
S7.1 Fig. Genome-wide association signals of the six methods in Bn. Based on the 40,087 candidate peak–gene pairs tested in Bn. See the shared legend for S7.1–S7.3 Figs.



S7.2 Fig. Genome-wide association signals of the six methods in CD4 T cells. Based on the 85,527 candidate peak–gene pairs tested in CD4 T cells. See the shared legend for S7.1–S7.3 Figs.



S7.3 Fig. Genome-wide association signals of the six methods in CD8 T cells. Based on the 50,931 candidate peak–gene pairs tested in CD8 T cells. See the shared legend for S7.1–S7.3 Figs.



S8 Fig. Overlap of significant links with the CIMA regulatory network in Bn and CD8 T cells. Odds ratio for a link significant for each method (within-gene $P_{\text{adj}} < 0.05$) being present in the CIMA SCENIC+ regulatory network, compared with links not in the network (2×2 table; all tested pairs in the cell type as background), as in Fig 3E for CD4 T cells. Light bars use eRegulons active in the cell type (cell-type eRegulons; 864 and 1,364 tested pairs in Bn and CD8 T cells), and dark bars the subset of these that belong to the high-quality eRegulons of the corresponding lineage (high-quality eRegulons; 817 and 382 tested pairs). Fisher's exact test $P < 10^{-30}$ for every bar; the dashed line marks an odds ratio of 1.

S1 Table. Pooled hypergeometric enrichment p-values across regulatory references.

Cell type	Method	EnhancerAtlas	FANTOM5	4DGenome	GTEx	HiChIP	PCHiC
Bn	Causal SMR	3.7×10^{-136}	3.9×10^{-100}	2.5×10^{-58}	2.3×10^{-82}	1.0×10^{-416}	3.4×10^{-58}
	Causal IVW	1.2×10^{-108}	1.5×10^{-88}	3.1×10^{-52}	1.6×10^{-71}	2.7×10^{-365}	2.6×10^{-46}
	Causal Egger	6.0×10^{-65}	8.5×10^{-68}	5.6×10^{-40}	4.2×10^{-38}	9.0×10^{-286}	1.1×10^{-31}
	Causal Joint	3.7×10^{-130}	1.1×10^{-99}	2.3×10^{-54}	4.2×10^{-77}	5.3×10^{-427}	9.3×10^{-57}
	Marginal Multiomics	6.1×10^{-138}	1.3×10^{-162}	5.1×10^{-61}	2.5×10^{-61}	2.1×10^{-879}	2.9×10^{-75}
	Marginal Pseudobulk	1.6×10^{-80}	2.6×10^{-94}	2.4×10^{-44}	8.4×10^{-55}	8.0×10^{-404}	4.0×10^{-54}
CD4 T cell	Causal SMR	3.5×10^{-302}	6.5×10^{-107}	9.4×10^{-50}	2.6×10^{-209}	1.3×10^{-911}	2.0×10^{-140}
	Causal IVW	1.2×10^{-221}	3.0×10^{-94}	3.2×10^{-47}	3.6×10^{-151}	5.0×10^{-890}	2.2×10^{-154}
	Causal Egger	2.1×10^{-109}	4.9×10^{-47}	5.7×10^{-33}	9.2×10^{-98}	3.0×10^{-584}	1.9×10^{-92}
	Causal Joint	6.0×10^{-249}	6.2×10^{-109}	1.1×10^{-55}	8.2×10^{-195}	4.7×10^{-981}	5.6×10^{-160}
	Marginal Multiomics	1.7×10^{-200}	3.2×10^{-123}	6.1×10^{-40}	2.3×10^{-114}	6.2×10^{-1875}	1.3×10^{-191}
	Marginal Pseudobulk	1.3×10^{-137}	4.7×10^{-92}	4.1×10^{-37}	5.3×10^{-97}	1.6×10^{-1172}	1.4×10^{-158}
CD8 T cell	Causal SMR	2.1×10^{-204}	1.8×10^{-72}	2.4×10^{-38}	1.7×10^{-104}	1.6×10^{-614}	8.6×10^{-103}
	Causal IVW	3.1×10^{-156}	1.5×10^{-60}	1.7×10^{-34}	8.7×10^{-100}	1.1×10^{-616}	6.9×10^{-83}
	Causal Egger	3.4×10^{-104}	3.1×10^{-38}	1.2×10^{-28}	6.4×10^{-55}	1.2×10^{-446}	7.1×10^{-48}
	Causal Joint	8.8×10^{-189}	1.4×10^{-76}	3.9×10^{-41}	1.5×10^{-119}	4.6×10^{-687}	4.2×10^{-95}
	Marginal Multiomics	1.7×10^{-150}	3.1×10^{-111}	2.1×10^{-34}	2.9×10^{-75}	7.4×10^{-1466}	4.1×10^{-130}
	Marginal Pseudobulk	1.6×10^{-104}	1.4×10^{-77}	3.6×10^{-17}	9.5×10^{-57}	1.0×10^{-824}	1.7×10^{-77}

For each cell type, method, and reference, an upper-tail hypergeometric test was applied to counts pooled over eligible genes. For each gene, N is the number of background candidate peaks, K the number overlapping the reference, n_{sig} the number of selected significant peaks, and k the selected peaks overlapping the reference. Eligibility requires $N > 0$, $K > 0$, $n_{\text{sig}} > 0$, and $N > K$. The test uses $\sum N, \sum K, \sum n_{\text{sig}}, \sum k$. Entries are hypergeometric p-values, with no additional adjustment across the displayed comparisons specified by the source workbook. Bn, naive B cells; CD4 T cell / CD8 T cell, naive $CD4^+$ / $CD8^+$ T cells. Causal Joint is the Cauchy combination (CCT) of SMR, IVW, and MR-Egger; Marginal Multiomics denotes the WNN metacell correlation and Marginal Pseudobulk the individual-level pseudobulk correlation.

S2 Table. Pooled regulatory-reference fold enrichment across cell types and methods.

Cell type	Method	EnhancerAtlas	FANTOM5	4DGenome	GTEx	HiChIP	PCHiC
Bn	Causal SMR	**3.59**	**6.18**	**5.00**	**2.96**	**4.46**	**2.70**
	Causal IVW	3.10	5.65	4.54	2.71	4.02	2.46
	Causal Egger	2.69	5.59	4.39	2.34	3.89	2.32
	Causal Joint	3.20	5.67	4.42	2.70	4.12	2.56
	Marginal Multiomics	1.98	3.56	2.61	1.70	3.08	1.84
	Marginal Pseudobulk	2.49	5.03	3.76	2.34	3.76	2.38
CD4 T cell	Causal SMR	**3.56**	**4.45**	**3.30**	**3.01**	**4.65**	2.55
	Causal IVW	2.98	4.08	3.12	2.63	4.41	**2.61**
	Causal Egger	2.52	3.46	3.02	2.51	4.17	2.41
	Causal Joint	2.99	4.17	3.20	2.77	4.40	2.56
	Marginal Multiomics	1.84	2.49	1.85	1.68	3.35	1.85
	Marginal Pseudobulk	2.05	3.10	2.30	1.94	3.78	2.16
CD8 T cell	Causal SMR	**3.65**	**4.64**	**3.60**	**2.76**	**5.46**	**2.70**
	Causal IVW	3.13	4.19	3.36	2.69	5.26	2.52
	Causal Egger	2.90	3.84	3.48	2.40	5.08	2.29
	Causal Joint	3.21	4.41	3.46	2.75	5.22	2.53
	Marginal Multiomics	1.90	2.81	2.00	1.68	4.01	1.88
	Marginal Pseudobulk	2.21	3.60	2.11	1.96	4.62	2.07

Fold enrichment is $(\sum k / \sum n_{\text{sig}})/(\sum K / \sum N)$, using the eligible-gene counts defined in S1 Table. Values are rounded to two decimal places. Bold indicates the greatest unrounded enrichment within each cell type and reference column. Contributing genes and selected-set sizes can differ across methods; these descriptive values are not paired tests of method superiority. Bn, naive B cells; CD4 T cell / CD8 T cell, naive $CD4^+$ / $CD8^+$ T cells. Causal Joint is the Cauchy combination (CCT) of SMR, IVW, and MR-Egger; Marginal Multiomics denotes the WNN metacell correlation and Marginal Pseudobulk the individual-level pseudobulk correlation.

S3 Table. Data overview for the three cell types analyzed.

	Bn	CD4 T cell	CD8 T cell
Single-cell data used for the pseudobulk correlation (after quality control)			
Individuals (passing RNA and ATAC cell-count filters)	335	372	346
scRNA-seq cells	160,577	916,643	460,439
scATAC-seq cells	123,580	360,343	210,077
scRNA-seq cells per individual, median (IQR)	436 (315–619)	2,336 (1,655–3,150)	1,200 (763–1,760)
scATAC-seq cells per individual, median (IQR)	348 (236–464)	898 (618–1,278)	510 (320–845)
Genes in the pseudobulk RNA matrix	36,326	36,326	36,326
Peaks in the pseudobulk ATAC matrix	338,036	338,036	338,036
cis-QTL summary statistics			
eQTL sample size	405	406	405
caQTL sample size	385	385	384
Peaks with at least one caQTL at $P < 10^{-8}$	3,449	6,110	3,847
Candidate peak–gene pairs and the Mendelian randomization analysis set			
Candidate gene–peak pairs (All_GenePeak, TSS–peak distance < 1 Mb)	3,814,826	5,146,441	4,505,251
With caQTL and eQTL variants in common (at least 3 shared variants)	46,190	109,961	62,136
Gene and peak present in both pseudobulk and WNN data	40,389	86,204	51,649
After removing pairs with a missing p-value in any base method	40,153	85,782	51,123
Final analysis set (IVW residual variance ≤ 10)	40,087	85,527	50,931
Genes / peaks in the final analysis set	8,149 / 2,951	10,701 / 5,338	9,427 / 3,327
Variants (instruments) per pair in the IVW fit, median (IQR)	22 (10–40)	22 (10–43)	22 (9–41)
Samples for the marginal correlation methods			
Pseudobulk correlation: individuals	335	372	346
WNN correlation: metacells (external PBMC multiome)	80	80	80

Bn, naive B cells (CIMA L4 type Bn_TCL1A); CD4 T cell, naive $CD4^+$ T cells (CD4_Tn_CCR7); CD8 T cell, naive $CD8^+$ T cells (CD8_Tn_CCR7). The Chinese Immune Multi-Omics Atlas (CIMA; Yin et al., Science 2026; OMIX009786) comprises scRNA-seq and scATAC-seq data from 428 Chinese adults annotated into 73 L4 immune cell types. The cis-QTL were mapped within 1 Mb of the gene transcription start site or the peak midpoint, using genotypes of the whole cohort (5,051,397 variants with minor allele frequency > 0.1); the numbers in the table are specific to the three cell types analyzed here. Three sample definitions are used: the QTL sample sizes (summary statistics used by the Mendelian randomization methods), the individuals in the pseudobulk matrices (those passing the RNA and ATAC cell-count filters; pseudobulk correlation), and 80 metacells of an external PBMC multiome data set (WNN correlation). The caQTL and eQTL data come from the same cohort, so the two QTL data sets overlap in samples. Peaks with at least one caQTL at $P < 10^{-8}$ are counted from the significant caQTL files used to select instruments; this is a nominal threshold, not the false-discovery-rate-defined caPeak count of CIMA. Candidate pairs are All_GenePeak pairs with at least 3 variants shared between the clumped caQTL and the eQTL data, whose gene and peak are present in both the pseudobulk and WNN data. Pairs with a missing p-value in any base method (IVW, MR-Egger, SMR, WNN, pseudobulk) or with an IVW residual variance above 10 were removed. IQR, interquartile range.

S4 Table. GWAS used in the S-LDSC analysis (Fig 4A) and in the GWAS-overlap tracks (Fig 4B).

Trait	Study or data source	N	SNPs	h^2 (SE)
Ankylosing spondylitis (AS)	Pan-UK Biobank (`https://pan.ukbb.broadinstitute.org/`)	420,473 (1,185 cases / 419,288 controls)	1,188,036	0.0052 (0.0017)
Crohn's disease (CD)	International IBD Genetics Consortium (`https://www.ibdgenetics.org/`)	367,592 (20,873 cases / 346,719 controls)	1,188,159	0.0649 (0.0058)
Inflammatory bowel disease (IBD)	de Lange et al. [4]	59,957	1,150,510	0.3202 (0.0243)
Rheumatoid arthritis (RA)	Saevarsdottir et al. [5]	1,026,690	1,184,671	0.0132 (0.0011)
Systemic lupus erythematosus (SLE)	Bentham et al. [6]	14,267	1,163,805	0.4567 (0.0957)
Multiple sclerosis (MS)	International Multiple Sclerosis Genetics Consortium [7]	27,148	450,159	0.0800 (0.0373)
Type 1 diabetes (T1D)	Chiou et al. [8]	511,056 (median; range 25,452–520,580)	1,192,414	0.0397 (0.0054)

All seven GWAS were analyzed in S-LDSC (Fig 4A). N and SNPs are those of the summary statistics read by S-LDSC (HapMap3 SNPs; variants with a missing Z or N are dropped); for AS and CD, the numbers of cases and controls are given. The sample size of the T1D summary statistics varies across variants (median and range given). h^2 is the observed-scale SNP heritability estimated by S-LDSC with the 97 baselineLD v2.2 annotations and the 1000 Genomes Phase 3 European LD reference (all-candidate-peak model; standard error in parentheses). The summary statistics for IBD, RA, SLE and T1D were obtained from the collection of 107 independent GWAS on Zenodo (doi:10.5281/zenodo.10515792), those for AS from the Pan-UK Biobank, those for CD from the International IBD Genetics Consortium website, and those for MS from ImmunoBase (protected data; approval from the International Multiple Sclerosis Genetics Consortium is required).

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